SYSDESIGN REFERENCE

Cache Invalidation Strategies

A senior-level reference with tradeoffs, failure modes, and interview follow-ups for cache invalidation strategies.

Define how cached representations become fresh enough after authoritative data changes.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / CACHE-INVALIDATION

SYSDESIGN

Cache Invalidation Strategies

CORE IDEA

Define how cached representations become fresh enough after authoritative data changes.

Reference architecture decision chain

01	02	03	04
Ttl	Event Invalidation	Versioned Keys	Write-Through

Decisions to defend

+ TTL
+ event invalidation
+ versioned keys
+ write-through
+ and stale tolerance.

Failure modes to test

+ Lost invalidations
+ broad purges
+ race conditions
+ and cascading misses.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Cache Invalidation Strategies

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS

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